

AAYUSH BHARADWAJ

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EDUCATION

University of Colorado Boulder

MS, Data Science — College of Engineering & Applied Science

Boulder, CO

Aug 2025 – May 2027

Coursework: NLP; Neural Networks & Deep Learning; Statistical Learning; Machine Learning; Data Mining; Data Structures & Algorithms; Statistical Methods I & II

Birla Institute of Technology, Mesra

Integrated MSc, Physics

Ranchi, India

2020 – 2025

Coursework: Quantum Mechanics I & II; Statistical Mechanics; Electrodynamics; Computational Physics; CFD; Numerical Methods; Probability & Statistics; Data Structures & Algorithms; C/C++

PROFESSIONAL EXPERIENCE

Honda 99P Labs — Capstone Project, University of Colorado Boulder

AI/ML Researcher — Agent Evaluation & Data Science | Fall 2026 (Aug 2026 – Present) | Boulder, CO

- Build and evaluate an AI agent on a real task; design scoring across quantitative metrics, qualitative judgment, and implicit signals (retries, rewrites, abandonment) to define good agent performance before production deployment.
- Construct a multi-labeler ground-truth pipeline with inter-rater agreement analysis; validate automated scoring harness output against human judgment for reliability and reproducibility.

PROJECTS & RESEARCH EXPERIENCE

NebulaSense LogIQ — Cloud-Native Anomaly Intelligence Engine

ML Engineer & Systems Designer — Personal Project | Jan 2026 | Boulder, CO

- Built an end-to-end anomaly detection platform: 2-layer LSTM autoencoder (AUC > 0.90) trained semi-supervised on distributed microservice logs; deployed via FastAPI REST inference service with Docker/Kubernetes manifests.

Micro — Adaptive AI Study Platform (HackCU 12)

Data Scientist & ML Engineer — HackCU 12 (MLH), CU Boulder | Mar – Apr 2026 | Boulder, CO

- Engineered a Bayesian mastery-modeling engine (Beta-Bernoulli framework) with full posterior inference, credible intervals, and confidence-/speed-weighted evidence accumulation; integrated Canvas API for context-aware navigation.

Will It Chart — Hit-Song Prediction Engine

ML Engineer — Personal Project | May 2026 | Boulder, CO

- Trained an XGBoost model on 114,000 Spotify tracks (ROC-AUC 0.73, 5-fold CV 0.725 ± 0.004); applied SHAP explainability and deployed a live Gradio dashboard with hit-probability gauge and per-feature breakdown.

GhostHunt — Fraudulent Job Posting Detector

ML Engineer — Personal Project | May 2026 | Boulder, CO

- Engineered 18 behavioral features + TF-IDF NLP features; trained an XGBoost classifier achieving 87% held-out accuracy; applied SHAP for per-prediction interpretability and deployed a live Gradio scoring interface.

BubbleBreak — Recommender Fairness Simulation

ML Engineer — Personal Project | June 2026 | Boulder, CO

- Built a 1,000-agent, 90-day recommendation simulation; trained a fairness-aware PyTorch model reducing filter-bubble formation by 34% at a 2.1% engagement cost.

GitHub Repository Popularity — Statistical Analysis

Statistical Researcher — STAT 4010/5010, CU Boulder | Spring 2026 | Boulder, CO

- Fit MLR and Negative Binomial GLM on repository popularity across 2 datasets ($R^2 = 0.61/0.76$; AIC 7,487 vs. Poisson 74,851); confirmed language effects via ANOVA/Kruskal-Wallis.

UXposed — Dark-Pattern Detection & Compliance Analytics

ML Engineer — Personal Project | Jul 2026 | Boulder, CO

- Scraped and classified UI copy across 50 Fortune 500 checkout flows into 8 FTC/EU DSA dark-pattern categories; built a severity-weighted Manipulation Index dashboard.

ADDITIONAL PROJECTS

Campus Event Scheduling System & Dijkstra's Algorithm

Boulder, CO

Conflict-free scheduling engine using min-heaps, balanced trees, and hash maps; Dijkstra shortest-path routing with full complexity analysis.

DrawQuery

Boulder, CO

Vision transformer converting hand-drawn ER diagrams into valid SQL DDL, sub-3-second via Gradio.

NFL Data Mining & Analytics

Boulder, CO

ETL pipelines and classification models for game-outcome prediction.

ClinsightAI

Boulder, CO

Healthcare decision intelligence platform with schema-agnostic ingestion and PCA-based clustering.

CrispAI

Boulder, CO

Real-time speech enhancement transformer trained on Microsoft's DNS Challenge dataset (PESQ 3.2).

SceneScript

Boulder, CO

CLIP + LLM pipeline generating timestamped natural-language video summaries.

NASA Exoplanet Statistical Analysis

Boulder, CO

EDA and bootstrap hypothesis testing on NASA's Exoplanet Archive dataset.

CMB Dipole Anisotropy & FIRAS Analysis (MSc Thesis)

Boulder, CO

Blackbody radiation inversion on COBE/FIRAS data; derived observer velocity of 408.5 km/s.

SKILLS, CERTIFICATIONS & HONORS

Technical: Python, R, SQL, C, C++, MATLAB, Pandas, NumPy, Scikit-learn, FastAPI, Streamlit, Gradio, Docker, Kubernetes, Git, REST API design, OOP, Data Structures & Algorithms, distributed systems

Machine Learning & Statistics: Bayesian inference, GLMs (Poisson, NB), MLR, ANOVA, hypothesis testing, bootstrapping, XGBoost, SHAP, LSTM autoencoders, PyTorch, recommender systems, cross-validation

Certifications & Honors: Azure MLOps & GenAI (certified); AWS Analytics & ML (certified); 1st, CFD Analysis, SAE ISS ADC; Co-Head, Astronomy & Astrophysics Club